

AI-based Urban Traffic Management System using Reinforcement Learning and Computer Vision

PRESENTED BY

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PROJECT GUIDE

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DOMAIN IDENTIFICATION

- **Domain:** Artificial Intelligence & Smart Transportation
- **Area:** Intelligent Traffic Management Systems
- **Technology:** Reinforcement Learning + Computer Vision
- **Application:** Smart City Traffic Signal Optimization



LITERATURE REVIEW



TITLE JOURNAL	AUTHOR	YEAR	CONTENT
Intelligent Traffic Signal Control Based on Reinforcement Learning with Edge Computing and Intelligent Reflecting Surface <i>Zhang, G., Yue, Y., Yang, J., & Tang, J. (2024) — IEEE Access</i>	Zhang, G., Yue, Y., Yang, J., & Tang, J.	2024	Combines reinforcement learning and edge computing for faster real-time traffic decisions. Supports intelligent and scalable traffic signal optimization.
Reinforcement Learning for Traffic Signal Control in Hybrid Action Space <i>Luo, H., & Jin, S. (2022) — IEEE Transactions on Intelligent Transportation Systems</i>	Luo, H., & Jin, S.	2022	Introduces an RL-based model that adapts signal timing in complex traffic scenarios. Helps reduce congestion and improve traffic management.
Learning Traffic Signal Control from Demonstrations <i>Zheng et al. (2020) — Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)</i>	Zheng et al.	2020	Proposes learning traffic signal policies from expert demonstrations. Achieves stable and efficient signal control with reduced manual intervention.
Deep Learning for Real-Time Traffic Signal Control <i>Chen Chen (2017) — IEEE Intelligent Transportation Systems Magazine</i>	Chen Chen	2017	Applies deep learning for dynamic traffic signal control using real-time data. Enhances efficiency and adaptability in urban traffic systems.
Traffic Signal Timing via Deep Reinforcement Learning <i>L. Li (2016) — IEEE Transactions on Intelligent Transportation Systems</i>	L. Li	2016	Uses deep reinforcement learning to optimize traffic signal timing based on traffic conditions. Improves traffic flow and reduces waiting time compared to fixed signals.

OBJECTIVE



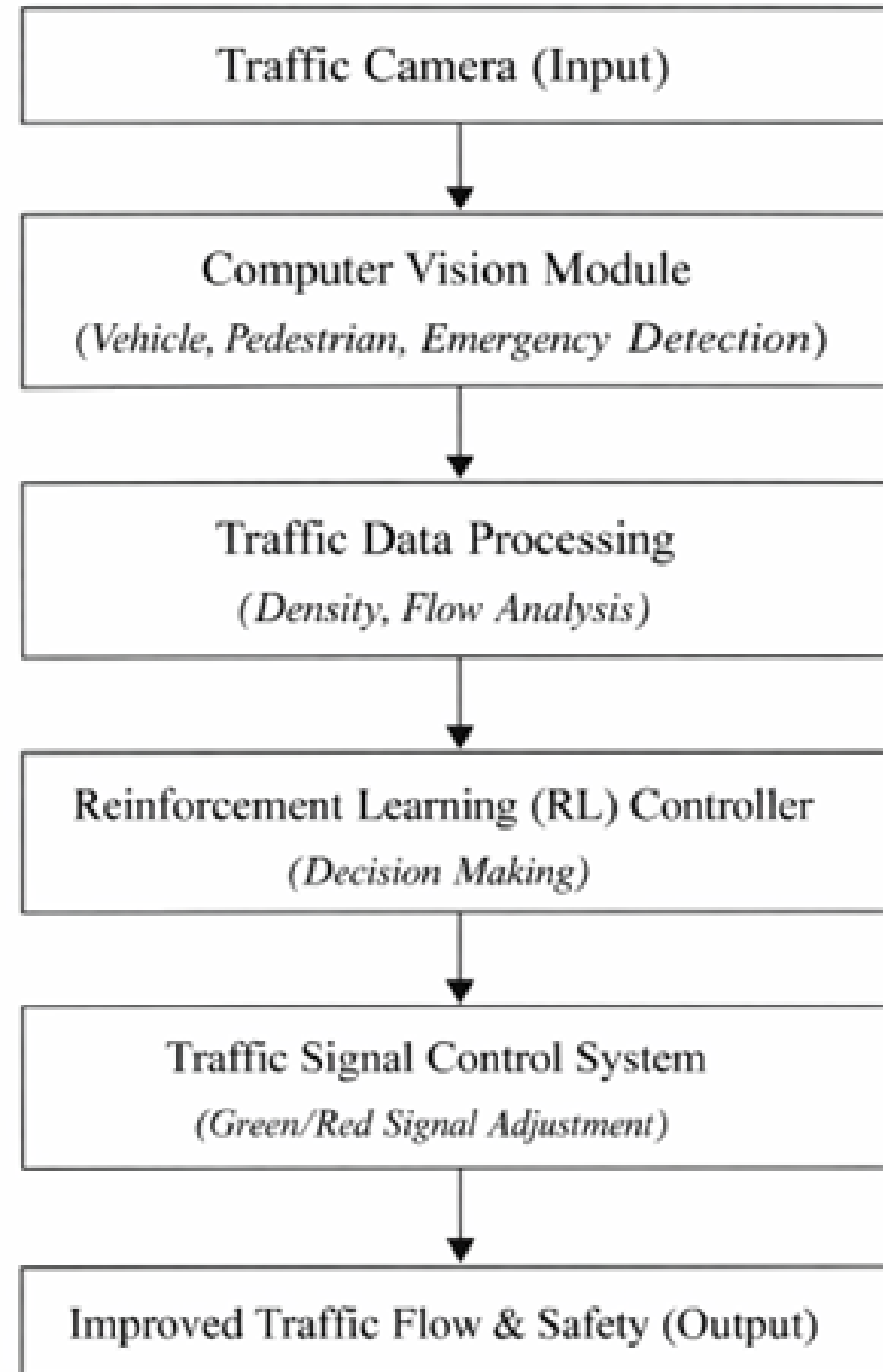
- Provide real-time emergency vehicle priority . Enhance pedestrian safety through vision monitoring
- Optimize traffic using real-time congestion and weather data hence reducing waiting time, fuel use, and energy consumption

PROJECT PROPOSAL

- Proposes an AI-based urban traffic management system.
- Uses real-time data to dynamically adjust traffic signals.
- Applies Computer Vision to detect vehicles, pedestrians, and emergency vehicles.
- Utilizes Reinforcement Learning to optimize signal timing decisions.
- Replaces traditional fixed-time traffic signals with adaptive control.
- Aims to improve traffic flow efficiency, road safety, and support smart city infrastructure.



BLOCK DIAGRAM



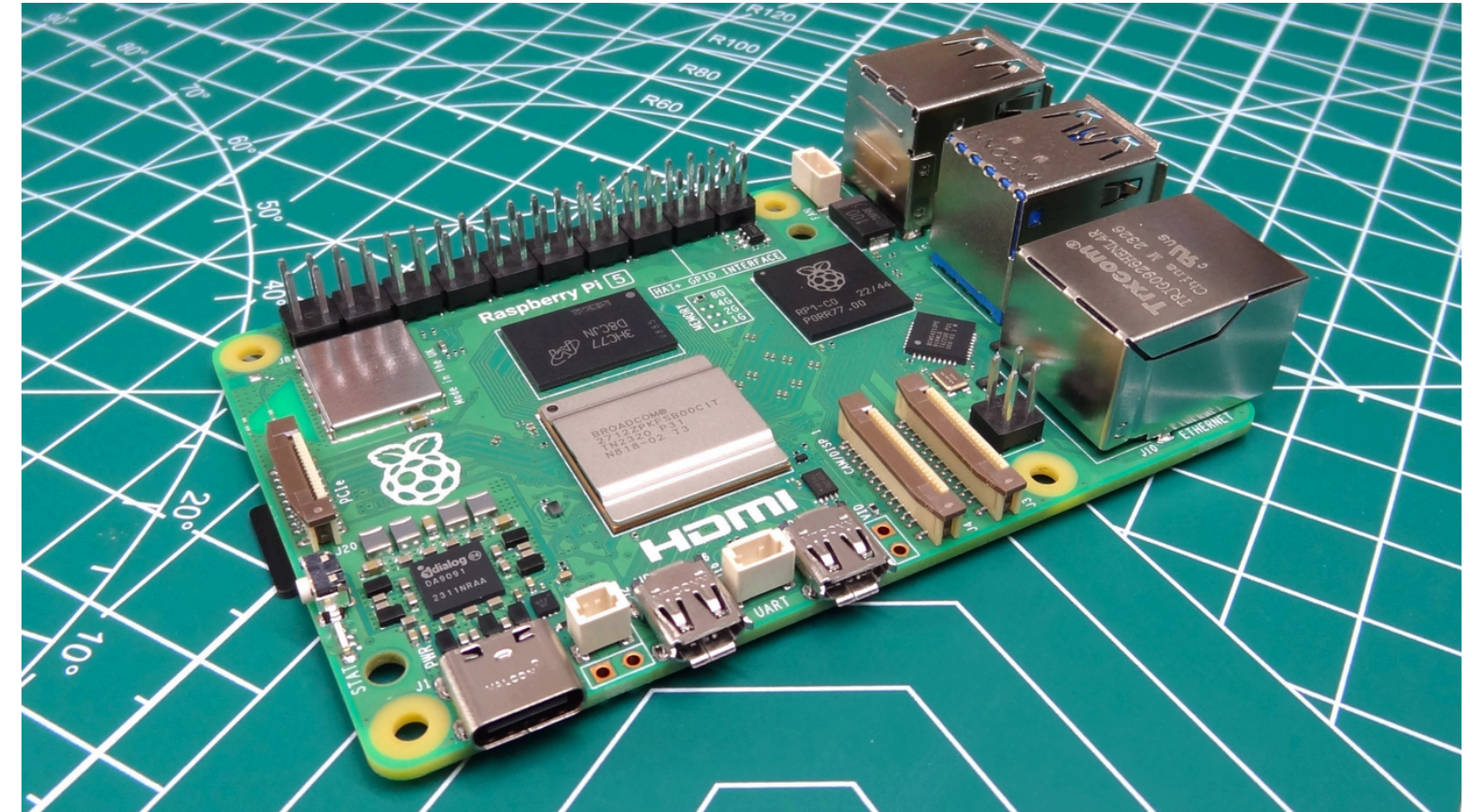
HARDWARE AND SOFTWARE USED

Software Components :

- Flask
- SQLite
- YOLOv8
- OpenCV
- TensorFlow (DQN)
- Raspberry Pi (RPi.GPIO)

Hardware Components :

- Raspberry Pi
- USB Camera
- IR Sensors (×4)
- Traffic Light Modules (×4)
- Jumper Wires
- Breadboard



PROGRESS

- **System Design & Architecture : Completed**
- **Work Distribution (Software + Hardware Split):**
 1. Software Development (Simulation) :332
 2. Hardware Preparation (IoT Setup) : ALL
 3. IoT Communication & Control Logic: 354
 4. Dashboard & Real-Time Monitoring: 347
- **Current Stage :**
 1. Software simulation working successfully.
(Software Upated after last review)
 1. Hardware module under implementation
and integration phase.(70% Completed)



INTERIM RESULTS

- **Improved Traffic Flow**
- **Reduction in Waiting Time**
- **Emergency Vehicle Performance**
- **Pedestrian Safety Improvement**
- **Learning Behaviour of RL Agent**
- **System Stability**
- **Current Performance Indicators (Preliminary):**
 1. Average waiting time reduced by approximately 20–30%
 2. Traffic throughput increased by approximately 10–15%
 3. Emergency vehicle delay reduced by approximately 35–40%
 4. Signal efficiency improving with continuous learning



THANK YOU!